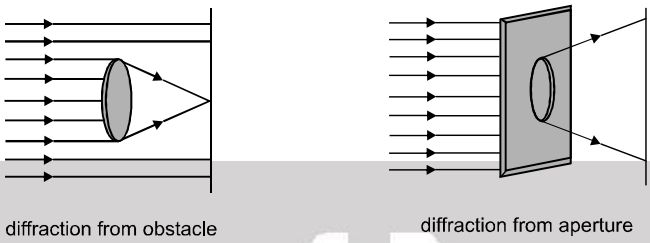
## DIFFRACTION OF LIGHT

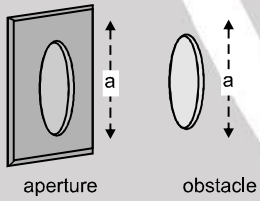
Bending of light rays from sharp edges of an opaque obstacle or aperture and its spreading in the eometricle shaddow region is defined as diffraction of light or deviation of light from its rectilinear propogation tendency is defined as diffraction of light.



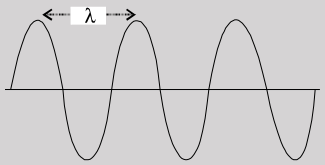

- Diffraction was discovered by Grimaldi
- Theoritically explained by Fresnel
- Diffraction is possible in all type of waves means in mechanical or electromagnetic waves shows diffraction.

  Diffraction depends on two factors :

  (i) Size of obstacles or aperture




  (ii) Wave length of the wave



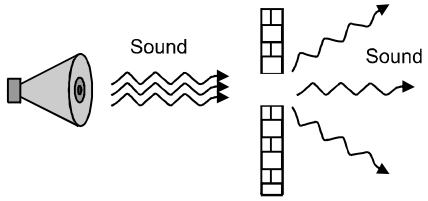

  Condition of diffraction. Size of obstacle or aperture should be nearly equal to the wave length of light

  $\lambda \simeq a$ $\qquad$ $\frac{a}{\lambda} \simeq 1$

  If size of obstacle is much greater then wave length of light, the rectilinear motion of light is observed.

- It is practically observed when size of aperture or obstacle is greater than 50 $\lambda$ then obstacle or aperture does not shows diffraction.
- Wave length of light is in the order $10^{-7}$m. In general obstacle of this wave length is not present so light rays does not show diffraction and it appears to travel in straight line Sound wave shows more diffraction as compare to light rays because wavelength of sound is high (16 mm to 16m). So it is generally diffracted by the objects in our daily life.

- Diffraction of ultrasonic wave is also not observed as easily as sound wave because their wavelength is of the order of about 1 cm. Diffraction of radio waves is very conveniently observed because of its very large wavelength (2.5 m to 250 m). X-ray can be diffracted easily by crystel. It was discovered by Lave.




diffraction of sound from a window

## TYPES OF DIFFRACTION

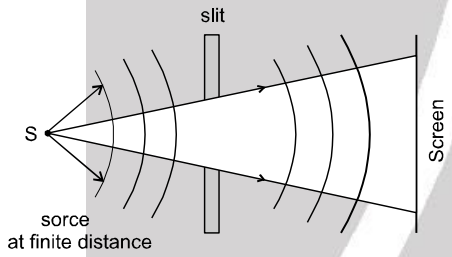
(i) There are two type of diffraction of light :

(a) Fresnel's diffraction (b) Fraunhofer's's diffraction

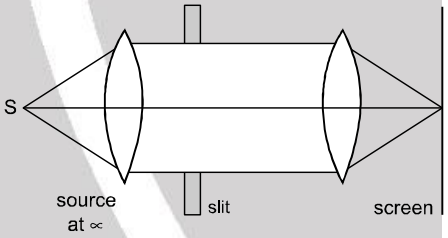
**(a) Fresnel diffraction**

If either source or screen or both are at finite from the diffracting device (obstacle or aperture), the diffraction is called fresnel diffraction and the pattern is the shadow of the diffracting device modified by diffraction effect.

Example :- Diffraction at a straight edge, small opaque disc, narrow wire are examples of fresnel diffraction.




Fresnel's diffraction




Fraunhofer's diffraction

**(b) Fraunhofer diffraction**

Fraunhofer diffraction is a particular limiting case of fresnel diffraction.

In this case, both source and screen are effectively at infinite distance from the diffracting device and pattern is the image of source modified by diffraction effects.

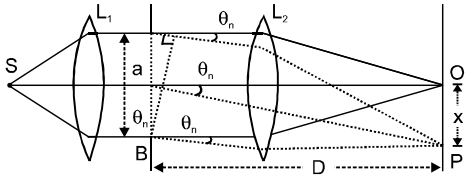
Example :- Diffraction at single slit, double slit and diffraction grating are the examples of fraunhofer diffraction.

**Comparison between fresnel and fraunhofer diffraction**

| | Fresnel Diffraction | Fraunhofer Diffraction |
|---|---|---|
| (a) | Source and screen both are at finite distance from the diffractor | Source and screen both are at infinite distance from the diffractor |
| (b) | Incident and diffracted wave fronts are spherical or cylinderical | Incident and diffracted wavefronts are plane due to infinite distance from source |
| (c) | Mirror or lennses are not used for obtaining the diffraction pattern | Lens are used in this diffraction pattern |
| (d) | Centre of diffraction pattern is sometime bright and sometime dark depending on size of diffractor and distance of observation point. | Centre of diffraction is always bright |
| (e) | Amplitude of wave coming from different half period zones are different due to difference of obliquity | Amplitude of waves coming from different half period zones are same due to same obliquity |

## FRAUNHOFER DIFFRACTION DUE TO SINGLE SLIT

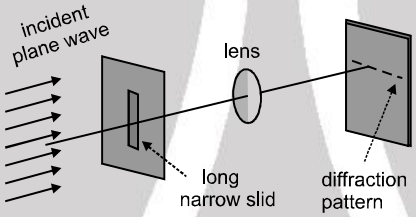
Ab is single slit of width a, Plane wavefront is incident on a slit AB. Secondary wavelets coming from every part of AB reach the axial point P in same phase forming the central maxima. The intensity of central maxima is maximum in this diffrection. Where $\theta_n$ represents direction of $n^{th}$ minima Path difference BB' = a sin $\theta_n$\




for $n^{th}$ minima a sin $\theta_n = n\lambda$ $\quad\therefore\quad$ $\sin\theta_n \approx \theta_n = \frac{n\lambda}{a}$ (if $\theta_n$ is small)

- When path difference between the secondary wavelets coming from A and B is $n\lambda$ or $2n\left[\frac{\lambda}{2}\right]$

  or even multiple of $\frac{\lambda}{2}$ then minima occurs




  For minima $\quad a\sin\theta_n = 2n\left[\frac{\lambda}{2}\right]$

- When path difference between the secondary wavelets coming from A and B is $(2n + 1)\frac{\lambda}{2}$

  or odd multiple of $\frac{\lambda}{2}$ then maxima occurs

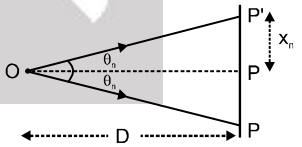
  For maxima $\quad a\sin\theta_n = (2n + 1)\frac{\lambda}{2}$ $\quad$ where n = 1, 2, 3 .....

  $n = 1 \rightarrow$ first maxima $\quad$ and $\quad$ $n = 2 \rightarrow$ second maxima.

- In alternate order minima and maxima occurs on both sides of central maxima.

**For $n^{th}$ minima**

If distance of $n^{th}$ minima from central maxima = $x_n$
distance of slit from screen = D, width of silt = a




Path difference $\delta = a\sin\theta_n = \frac{2n\lambda}{2} \Rightarrow \sin\theta_n = \frac{n\lambda}{a}$

In $\Delta$ POP' $\tan\theta_n = \frac{x_n}{D}$ $\quad$ If $\theta_n$ is small $\Rightarrow \sin\theta_n \approx \tan\theta_n \approx \theta_n$

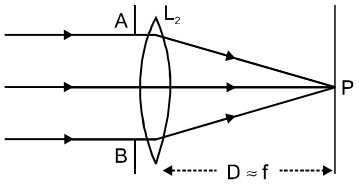
$X_n = \frac{n\lambda D}{a} \quad \Rightarrow \quad \theta_n = \frac{x_n}{D} = \frac{n\lambda}{a}$ First minima occurs both sides on central maxima.

For first minima $x = \frac{D\lambda}{a}$ $\quad$ and $\quad$ $\theta = \frac{x}{D} = \frac{\lambda}{a}$

- Linear width of central maxima $w_x = 2x \Rightarrow w_x = \frac{2d\lambda}{a}$
- Angular width of central maxima $w_\theta = 2\theta = \frac{2\lambda}{a}$

## SPECIAL CASE

Lens $L_2$ is shifted very near to slit AB.
In this case distance between slit and screen will be nearly equal to the focal lenght of lense $L_2$
(i.e. $D \approx f$)




$\theta_n = \frac{x_n}{f} = \frac{n\lambda}{a} \quad \Rightarrow \quad x_n = \frac{n\lambda f}{a}$

$w_x = \frac{2\lambda f}{a}$

and angular width of cnetral maxima $w_B = \frac{2x}{f} = \frac{2x}{a}$

**Fringe width**

Distance between two consecutive maxima (bright fringe) or minima (dark fringe) is known as fringe width.

$\beta = x_{n+1} - x_n = (n+1)\frac{\lambda D}{a} - \frac{n\lambda D}{a} = \frac{\lambda D}{a}$

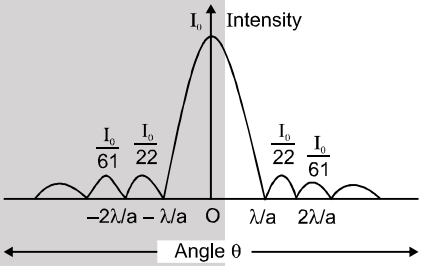
**Intensity curve of Fraunhofer's diffraction**

Intensity of maxima in Fraunhofer's diffrection is determined by $I = \left[\frac{2}{(2n+1)\pi}\right]^2 I_0$

$I_0$ = intensity of central maxima
n = order of maxima

intensity of first maxima $I_1 = d\theta \frac{4}{9\pi^2} I_0 \approx \frac{I_0}{22}$

intensity of second maxima $I_2 = \frac{4}{25\pi^2} I_0 \approx \frac{I_0}{61}$




- Diffraction occurs in slit is always fraunhofer diffraction as diffraction pattern obtained from the cracks between the fingers, when viewed a distant tubelight and in YDSE experiment are fraunhofer diffraction

## GOLDEN KEY POINTS

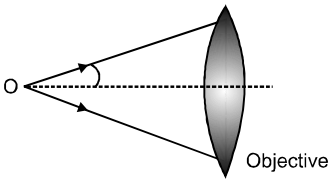
- The width of central maxima $\propto \lambda$, that is, more for red colour and less for blue.
  i.e., $w_x \propto \lambda$
  as $\lambda_{blue} < \lambda_{red} \Rightarrow w_{blue} < w_{red}$
- For obtaining the fraunhofer diffraction, focal length of second lens ($L_2$) is used.
  $w_x \propto \lambda \propto f \propto 1/a$
  width will be more for narrow slit
- By decreasing linear width of slit, the width of central maxima increase.

## RESOLVING POWER (R.P.)

A large number of images are formed as consequence of light diffraction of from a source. If two sources are separated such that their central maxima do not overlap, their images can be distinguished and are said to be resolved R.P. of an optical instrument is its ability to distinguish two neighbouring points.

Linear R.P. $d / \lambda D$ here D = Observed distance
Angular R.P. $d / \lambda$ d = Distance between two points,

**(1) Microscope :** In reference to a microscope, the minimum distance between two lines at which they are just distinct is called Resolving limit (RL) and it's reciprocal is called Resolving power (RPO)




$$\text{R.L.} = \frac{\lambda}{2\mu \sin\theta} \text{ and R.P.} = \frac{2\mu\sin\theta}{\lambda} \Rightarrow \text{R.P.} \propto \frac{I}{\lambda}$$

$\lambda$ = Wavelength of light used to illuminate the object

$\mu$ = Refractive index of the medium between object and objective.

$\theta$ = Half angle of the cone of light from the point object, $\mu \sin\theta$ = Numerical aperture.

**(2) Telescope :** Smallest angular separations ($d\theta$) between two distant object, whose images are separated in the telescope is called resolving limit. So resolving limit $d\theta = \frac{1.22\lambda}{a}$ and resolving power

$$(\text{RP}) = \frac{1}{d\theta} = \frac{a}{1.22\lambda} \Rightarrow \text{R.P.} \propto \frac{1}{2}$$ where a = aperture of objective.

## DIFFERENCE BETWEEN INTERFERENCE AND DIFFRACTION :

| | Interference | | Diffraction |
|---|---|---|---|
| (1) | It is the phenomenon of superposition of two waves coming from two different coherent sources | (1) | It is the phenomenon of superposition of two waves coming from two different parts of the same wave front. |
| (2) | In intrference pattern, all bright lines are equally bright and equally spaced | (2) | All bright lines are not equally bright and equally wide. Brightness and width goes on decreasing with the angle of diffraction. |
| (3) | All dark lines are totaally dark | (3) | Dark lines are not perfectly dark. Their contrast with bright lines and width goes on decreasing with angle of diffraction. |
| (4) | In interference bands are large in number | (4) | In diffraction bands are a few in number |

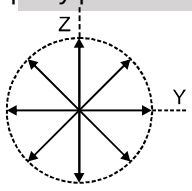
# POLARISATION

Experiments on interference and diffraction have shown that light is a form of wave motion. These effects do not tell us about the type of wave motion i.e. whether the light waves are longitudinal or transverese. The phenomenon of polarization has helped to establish beyond doubt that light waves are transverse waves.

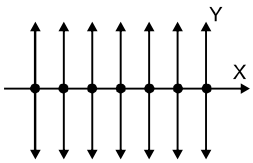
## UNPOLARISED LIGHT

An ordinary beam of light consists of a large number of waves emitted by the atoms of the light source.

Each atom produces a wave with its own orientation of electric vector $\vec{E}$ so all direction of vibration of $\vec{E}$ are equally probable.




unpolarised light propagating along. X-axis




unpolarised light

The resultant electromagnetic wave is a super position of waves produced by the individual atomic sources and it is called unpolarised light. In ordinary or unpolarised light, the vibrations of the electric vector occur symmetrically in all possible directions in a plane perpendicular to the direction of propagation of light.

## POLARISATION

The phenomenon of restricting the vibration of light (electric vector) in a particular direction perpendicular to the direction of propagation of wave is called polarisation of light.

In polarised light, the vibration of the electric vector occur in a plane perpendicular to the direction of propagation of light and are confined to a single direction in the plane (do not occur symmetrically in all possible directions.)

After polarisation the vibrations become asymmetrical about the direction of propagation of light.

## POLARISER

**Tourmaline crystal**

When light is passed through a tomaline crystal cut parallel to its optic axis, the vibrations of the light carrying out of the tourmaline crystal are confined only to one direction in a plane perpendicular to the direction of propagation of light. The emergent light from the crystal is said to be plane polarised light.

**Nicol Prism**

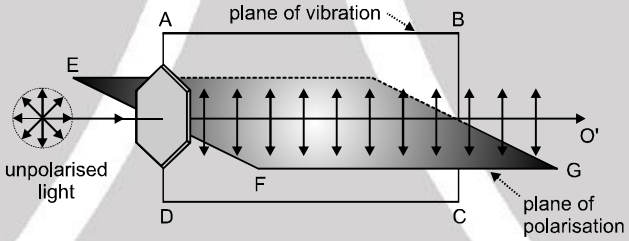
A nicol prism is an optical device which can be used for the production and detection of plane polarised light. It was invented by William Nicol in 1828.

**Polaroid**

A polaroid is a thin commercial sheet in the form of circular disc which makes use of the property of selective absorption to produce an intense beam of plane polarised light.

**PLANE OF POLARISATION AND PLANE OF VIBRATION :**

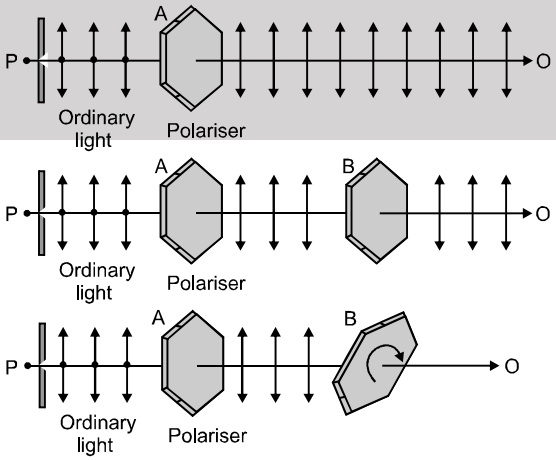
The plane in which vibrations of light vector and the direction of propogation lie is known as plane of vibration A plane normal to the plane of vibration and in which no vibration takes place is known as plane of polarisation.




**EXPERIMENTAL DEMONSTRATION OF POLARISATION OF LIGHT**

Take two tourmaline crystals cut parallel to their crystallographic axis (optic axis)

First hold the crystal A normally to the path of a beam of colour light. The emergent beam will be slightly coloured.




Rotate the crystal A about PO. No change in the intensity or the colour of the emergent beam of light. Take another crystal B and hold it in the path of the emergent beam of so that its axis is parallel to the axis of the crystal A. The beam of light passes through both the crystals and outcoming light appears coloured.

Now, rotate the crystal B about the axis PO. It will be seen that the intensity of the emergent beam decreases and when the axes of both the crystals are at right angles to each other no light comes out of the crystal B.

If the crystal B is further rotated light reappears and intensity becomes maximum again when their axes are parallel. This occurs after a further rotation of B through 90º

This experiment confirms that the light waves are transverse in nature.

The vibrations in light waves are perpendicular to the direction of propogation of the wave.

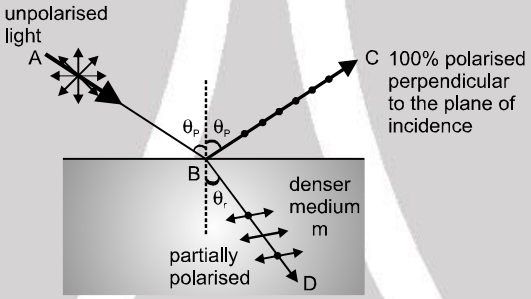
First crystal A polarises the light so it is called polariser.

Second crystal B, analyses the light whether it is polarised or not, so it is called analyser.

## METHODS OF OBTAINING PLANE POLARISED LIGHT

### Polarisation of reflection

The simplest method to produce plane polarised light is by reflection. This method was discovered by Malus in 1808. When a beam of ordinary light is reflected from a surface, the reflected light is partially polarised. The degree of polarisation of the polarised light in the reflected beam is greatest when it is incident at an angle called polarising angle or Brewster's angle.




### Polarising angle

Polarising angle is that angle of incidence at which the reflected light is completely plane polarisation.

### Brewster's Law

When unpolarised light strikes at polarising angle $\theta_P$ on a interface separating a rare medium from a denser medium of refractive index $\mu$, such that $\mu = \tan\theta_P$ then the reflected light (light in rare medium) is completely polarised. Also reflected and refracted and refracted rays are normal to each other.

This relation is known as Brewster's Law.

The law state that the tangent of the polarising angle of incidence of a transparent medium is equal to tis refractive index $\mu = \tan\theta_P$

In case of polarisation by reflection :

(i) For $i = \theta_P$ refracted light is partially polarised.

(ii) For $i = \theta_P$ reflected and refracted rays are perpendicular to each other.

(iii) For $i < \theta_P$ or $i > \theta_P$ both reflected and refracted light become partially polarised.

According to snell's law $\mu = \dfrac{\sin\theta_P}{\sin\theta_r}$ .....(i)

But according to Brewster's law $\mu = \tan\theta_O = \dfrac{\sin\theta_P}{\cos\theta_P}$ .....(ii)

From equation (i) and (ii) $\dfrac{\sin\theta_P}{\sin\theta_r} = \dfrac{\sin\theta_P}{\cos\theta_P} \Rightarrow \sin\theta_r = \cos\theta_P$

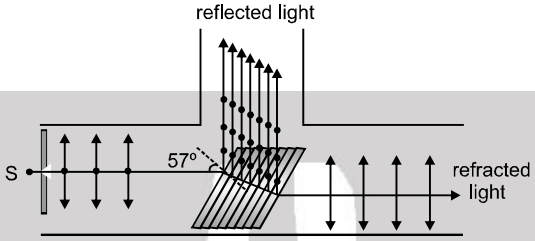
$\therefore$ $\sin\theta_r = \sin(90^o - \theta_P) \Rightarrow \theta_r = 90^o - \theta_P$ or $\theta_P + \theta_r = 90^o$

Thus reflected and refracted rays are mutually perpendicular

**By Refraction**

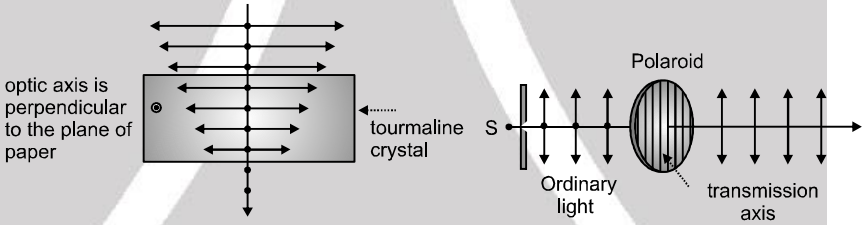
In this method, a pile of glass plates is formed by taking 20 to 30 microscope slides and light is made to be incident a polarising angle 57º. According Brewster law, the reflected light will be plane polarised with vibrations perpendicualr to the plane of incidence and the transmitted light will be partially polarised.

Since in one reflection about 15% of the light with vibration perpendicular to plane of paper is reflected therefore after passing through a number of plates emerging light will become plane polarised with vibrations in the plane of paper.



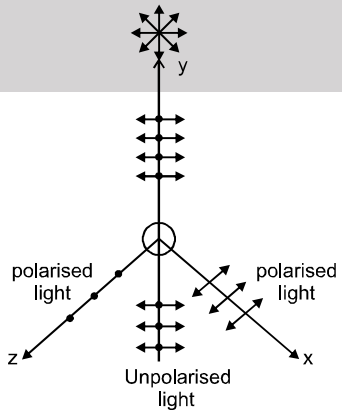

**By Dichroism**

Some crystals such as tourmaline and sheets of iodosulphate of quinone have the property of strongly absorbing the light with vibrations perpendicular of a specific direction (called transmision axis) and transmitting the light with vibration parallel to it. This selective absorption of light is called dichroism. So if unpolarised light passes through proper thickness of these, the transmitted light will plane polarised withvibrations parallel to transmission axis. Polaroids work onthis principle.



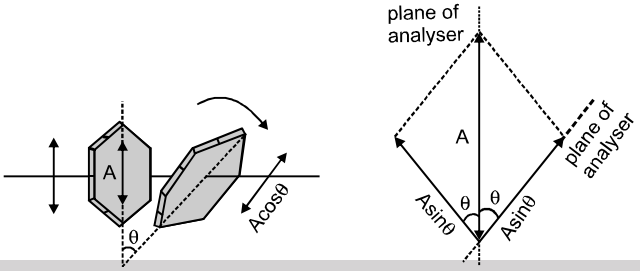

**By scattering :**

When light is incident on small particles of dust, air molecule etc. (having smaller size as compared to the wavelength of light,) it is absorbed by the electrons and is re-radiated in all directions. The phenomenon is called as scattering. Light scattered in a direction at right angles to the incident light is always plane-polarised.



**Law of Malus**

When a completely plane polarised light beam light beam is incident analyser, then intensity intensity of emergent light varies as the square of cosine of the angle between the planes of transmision of the analyser and the polarizer. $I \propto \cos^2\theta \Rightarrow I = I_0 \cos^2\theta$




(i) If $\theta = 0^o$ then $I = I_0$ maximum value (Parallel arrangement)

(ii) If $\theta = 90^o$ then, $I = 0$ minimum value (Crossed arrangement)

If plane polarised light of intensity $I_0$ (= $KA^2$) is incident on a polaroid and its vibrations of amplitude A make angle $\theta$ with transmission axis, then the component of vibrations parallel to transmission axis will be A cos$\theta$ while perpendicular to it will be be A sin$\theta$ .

Polaroid will pass only those vibrations which are parallel to transmission axis i.e. Acos $\theta$,

$\because$ $I_0 \propto A^2$

So the intensity of emergent light $I = K (A\cos\theta)^2 = KA^2\cos^2\theta$

If an unpolarised light is converted into plane polarised light its intensity becomes half.

If light of light emerging from the second polaroid is :

$I_2 = I_1 \cos^2\theta$ $\theta$ = angle between the transmission axis of the two polaroids.

**Optical Activity**

When plane polarised light passes through certain substances, the plane of polarisation of the emergent light is rotated about the direction of propagation of light through a certain angle. This phenomenon is optical rotation.

The substance which rotate the plane of polarision rotates the plane of polarisation is known as optical active substance. Ex. Sugar solution, sugar crystal, solidum chlorate etc.

Optical activity of a substance is measured with the help of polarimeter in terms of specific rotation which is defined as the rotaion produced by a solution of length 10 cm (1dm) and of unit concentration (1g/cc) for a given wave length of light at a given temp.

specific rotation $\left[\alpha\right]_{t^oC}^{\lambda} = \dfrac{\theta}{L \times C}$ $\theta$ = rotation in length L at concentration

**Types of optically active substances**

**(a) Dextro rotatory substances**

Those substance which rotate the plane of polarisation in clockwise direction are called dextro rotatory of right handed substances.

**(b) Laveo rotatory substances**

These substance which rotate the plane of polarisation in the anticlockwise direction are calledlaveo rotatory or left handed substances.

The amount of optical rotation depends upon the thickness and density of the crsytal or concentration in case of solutions, the temperature and the wavelength of light used.

Rotation varies inversely as the square of the wavelength of light.

## APPLICATIONS AND USES OF POLARISATION

- By determining the polarising and and using Brewster's law $\mu = \tan\theta_P$ refractive index of dark transparent substance can be determined.
- In calculators and watches, numbers and letters are formed by liquid crystals through polarisation of light called liquid crystal display (L.C.D)
- In CD player polarised laser beam acts as needle for producing sound from compact disc.
- It has also been used in recording and reproducing three dimensional pictures.
- Polarised light is used in optical stress analysis known as photoelasticity.
- Polarisation is also used to study asymmetries in molecules and crystals through the phenomenon of optical activity.

## Some Questions with their solutions

**1.** Light of wavelength 6000Å is incident normally on a slit of width $24 \times 10^{-5}$ cm. Find out the angular position of second minimum from central maximum ?

**2.** Light of wavelength 6328Å is incident normally on a slit of width 0.2 mm. Calculate the angular width of central maximum on a screen distance 9 m ?

**3.** Light of wavelength 5000Å is incident on a slit of width 0.1 mm. Find out the width of the central bright line on a screen distance 2m from the slit ?

**4.** The fraunhofer diffraction pattern of single slit is formed at the focal plane of a lens of focal length 1m. The width of the slit is 0.3 mm. If the thrid minimum is formed at a distance of 5 mm from the central maximum then calculate the wavelength of light.

**5.** Find the half angular width of the central bright maximum in the Fraunhofer diffraction pattern of a slit of width $12 \times 10^{-5}$ cm when the slit is illuminated by monochromatic light of wavelength 6000Å.

**6.** Light of wavelength 6000 Å is incident on a slit of width 0.30 mm. The screen is placed 2 m from the slit. Find (a) the position of the first dark fringe and (b) the width of the central bright fringe.

**7.** A Slit of width a is illuminated by monochromatic light of wavelength 650 nm at normal incidence. Calculate the value of a when :
(a) the first minimum falls at an angle of diffraction of 30º
(b) the first maximum falls at an angles of diffraction of 30º

**8.** Red light of wavelength 6500 Å from a distance source falls on a slit 0.50 mm wide. What is the distance between the first two dark bands on each side of the central bright of the diffraction pattern observed on a screen placed 1.8 m. from the slit.

**9.** In a single slit diffraction experiment first minimum for $\lambda_1$ = 660 nm coincides with first maxima for wavelength $\lambda_2$. Calculate $\lambda_2$.

**10.** Two polaroids are crossed to each other. When one of them is rotated through 60º, then what percentage of the incident unpolarised light will be transmitted by the polaroids ?

**11.** At what angle of incidence will the light reflected from water ($\mu$ = 1.3) be completely polarised ?

**12.** If light beam is incident at polarising angle (56.3º) on air-glass interface, then what is the angle of refraction in glass ?

**13.** A polariser and an analyser are oriented so that maximum light is transmitted, what will be the intensity of outcoming light when analyer is rotated through 60º.

# Solutions

**Sol.1** $a\sin\theta = 2\lambda$ given $\lambda = 6 \times 10^{-7}$ m, $a = 24 \times 10^{-5} \times 10^{-2}$ m

$\sin\theta = \frac{2\lambda}{a} = \frac{2\times 6\times 10^{-7}}{24\times 10^{-7}}$ $\therefore$ $\theta = 30^o$

**Sol.2.** given $\lambda = 6.328 \times 10^{-7}$ m, $a = 0.2 \times 10^{-3}$ m

$w_\theta = \frac{2\lambda}{a} = \frac{2\times 6.328\times 10^{-7}}{2\times 10^{-4}}$ radian $= \frac{6.328\times 10^{-3}\times 180}{3.14} = 0.36\equiv$

**Sol.3** $w_x = \frac{2\lambda}{a} = \frac{2\times 2\times 5\times 10^{-7}}{10^{-4}} = 20$ mm

**Sol.4** $x_n = \frac{nf\lambda}{a} \Rightarrow \lambda = \frac{ax_n}{fn} = \frac{3\times 10^{-4}\times 5\times 10^{-3}}{3\times 1} = 5000Å$ [$\because$ n = 3]

**Sol.5** $\because$ $\sin\theta = \frac{\lambda}{a}$ $\theta$ = half angular width of the central maximum.

$a = 12 \times 10^{-5}$ cm, $\lambda = 6000$ Å $= 6 \times 10^{-5}$ cm $\therefore \sin\theta = \frac{\lambda}{a} = \frac{6\times 10^{-5}}{12\times 10^{-5}} = 0.50 \Rightarrow \theta = 30^o$

**Sol.6** The first fringe is on either side of the central bright fringe.

here $n = \pm 1$, D = 2m, $\lambda = 6000$ Å $= 6$ j $10^{-7}$ m

$\therefore$ $\sin\theta = \frac{x}{D} \Rightarrow a = 0.30$ mm $= 3 \times 10^{-4}$ m $\Rightarrow a\sin\theta = n\lambda \Rightarrow \frac{ax}{D} = n\lambda$

(a) $x = \frac{n\lambda D}{a} \Rightarrow x = \pm\left[\frac{1\times 6\times 10^{-7}\times 2}{3\times 10^{-4}}\right] = \pm 4 \times 10^{-3}$ m

The positive and negative signs corresponds to the dark fringes on either side of the central bright fringe.

(b) The width of the central bright fringe $y = 2x = 2 \times 4 \times 10^{-3} = 8 \times 10^{-3}$ m = 8 mm

**Sol.7** (a) for first minimum $\sin\theta_1 = \frac{\lambda}{a}$

$\therefore$ $a = \frac{\lambda}{\sin\theta_1} = \frac{650\times 10^{-9}}{\sin 30^o} = \frac{650\times 10^{-9}}{0.5} = 1.3 \times 10^{-6}$ m

(b) for first maximum $\sin\theta_1 = \frac{3\lambda}{2a}$, $\therefore$ $a = \frac{3\lambda}{2\sin\theta} = \frac{3\times 650\times 10^{-9}}{2\times 0.5} = 1.95 \times 10^{-6}$ m

**Sol.8** Given $\lambda = 6500$Å $= 65 \times 10^{-8}$ m, $a = 0.5$ mm $= 0.5 \times 10^{-3}$ m., D = 1.8 mm

**Sol.9** For minima in diffraction pattern $d\sin\theta = n\lambda$

For first minima $d\sin\theta_1 = (1)\lambda_1 \Rightarrow \sin\theta_1 = \frac{\lambda_1}{d}$

for first maxima $d\sin\theta_2 = \frac{3}{2}\lambda_2 \Rightarrow \sin\theta_2 = \frac{3\lambda_2}{2d}$

Two will coincide if, $\theta_1 = \theta_2$ or $\sin\theta_1 = \sin\theta_2$

$\therefore$ $\frac{\lambda_1}{d} = \frac{3\lambda_2}{2d} \Rightarrow \lambda_2 = \frac{2}{3}\lambda_1 = \frac{2}{3} \times 660$ nm $= 440$ nm.

**Sol.10** Initially the polaroids are crossed to each other, that is the angle between their polarising directions is 90º. When one is rotated through 60º, then the angle between their polarising directions directions will become 30º.

Let the intensity of the incident unpolarised light = $I_0$

This light is plane polarised and passes through the second polaroid.

The intensity of light emerging form the second polaroid is $I_2 = I_1 \cos^2\theta$

$\theta$ = the angle between the polarising directions of the two polaroids.

$I_1 = \frac{1}{2} I_0$ and $\theta = 30^\circ$ so $O_2 = I_1 \cos^2 30^\circ \Rightarrow \frac{I_2}{I_0} = \frac{3}{8}$

$\therefore$ transmission percentage $= \frac{I_2}{I_0} \times 100 = \frac{3}{8} \times 100 = 37.5\%$

**Sol.11** $\mu = 1.3$, From Brewster's law $\tan \theta_P = \mu = 1.3 \quad \Rightarrow \theta = \tan^{-1} 1.3 = 53^\circ$

**Sol.12** $\because i_P + r_P = 90^\circ \quad \therefore r_P = 90^\circ - i_P = 90^\circ - 56.3^\circ = 33.7^\circ$

**Sol.13** According to Malus Law $I = I = I_0 \cos^2\theta = I_0 \cos^2 60^\circ = I_0 \left[\frac{1}{2}\right]^2 = \frac{I_0}{4}$